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FUEL PUMP WITH BUILT-IN THERMAL BYPASS

Abstract

A fuel delivery system for a gas turbine engine includes a fuel source, a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine, a main fuel filter positioned upstream of the main fuel pump, and an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

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Background/Summary

BACKGROUND

[0001] Exemplary embodiments pertain to the art of gas turbine engines, and more particularly to

fuel delivery systems of gas turbine engines.

[0002] Fuel delivery systems for gas turbine engines utilize a variety of pumps, often driven by a gearbox that extracts power from operation of the gas turbine engine to drive the pumps. The pumps include main pumps and an actuation pump. Modern aircraft require more efficient pumping systems and higher running fuel temperatures to improve overall thrust specific fuel consumption (TSFC), which is one measure of operational efficiency of the gas turbine engine and the aircraft.

[0003] Isolating the actuation pump from the high temperature main fuel flow allows for the use of a lighter, more efficient pump that requires a lower level of power extraction, but isolating the actuation pump also requires a cooling loop to prevent thermal runaway of the actuation pump.

BRIEF DESCRIPTION

[0004] In one exemplary embodiment, a fuel delivery system for a gas turbine engine includes a fuel source, a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine, a main fuel filter positioned upstream of the main fuel pump, and an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

[0005] Additionally or alternatively, in this or other embodiments a boost pump is positioned upstream of the main fuel filter and the actuation pump.

[0006] Additionally or alternatively, in this or other embodiments a wash filter is positioned fluidly between the boost pump and the main fuel filter.

[0007] Additionally or alternatively, in this or other embodiments the actuation pump is fed from the wash filter.

[0008] Additionally or alternatively, in this or other embodiments the second flow of fuel is urged from the one or more actuation devices to an outlet of the boost pump.

[0009] Additionally or alternatively, in this or other embodiments a thermal bypass valve is located downstream of the actuation pump and configured when activated to increase fuel flow of the second flow of fuel across the actuation pump when a temperature of the second flow of fuel exceeds a predetermined threshold, thereby decreasing the temperature of the second flow of fuel.

[0010] Additionally or alternatively, in this or other embodiments the thermal bypass valve is one of actively controlled or thermostatically controlled.

[0011] Additionally or alternatively, in this or other embodiments when the thermal bypass valve is activated the second flow of fuel is directed to the main fuel filter.

[0012] Additionally or alternatively, in this or other embodiments a high pressure relief valve is positioned downstream of the actuation pump and is configured when activated to direct the second flow of fuel toward the main fuel filter when a pressure of the second flow of fuel exceeds a predetermined threshold.

[0013] Additionally or alternatively, in this or other embodiments the actuation pump is a variable displacement pump.

[0014] In another exemplary embodiment, a gas turbine engine and fuel delivery system includes a gas turbine engine, including a turbine, and a combustor where a first flow of fuel is combusted to drive the turbine via a flow of combustion products. A fuel delivery system operably is connected to the gas turbine engine and includes a fuel source, a main fuel pump configured to deliver a first flow of fuel to the combustor, a main fuel filter positioned upstream of the main fuel pump, and an actuation pump positioned upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

[0015] Additionally or alternatively, in this or other embodiments a boost pump is positioned upstream of the main fuel filter and the actuation pump.

[0016] Additionally or alternatively, in this or other embodiments a wash filter positioned fluidly between the boost pump and the main fuel filter.

[0017] Additionally or alternatively, in this or other embodiments the actuation pump is fed from the wash filter.

[0018] Additionally or alternatively, in this or other embodiments the second flow of fuel is urged from the one or more actuation devices to an outlet of the boost pump.

[0019] Additionally or alternatively, in this or other embodiments a thermal bypass valve is located downstream of the actuation pump and configured when activated to increase fuel flow of the second flow of fuel across the actuation pump when a temperature of the second flow of fuel exceeds a predetermined threshold, thereby decreasing the temperature of the second flow of fuel.

[0020] Additionally or alternatively, in this or other embodiments the thermal bypass valve is one of actively controlled or thermostatically controlled.

[0021] Additionally or alternatively, in this or other embodiments when the thermal bypass valve is activated the second flow of fuel is directed to the main fuel filter.

[0022] Additionally or alternatively, in this or other embodiments a high pressure relief valve is positioned downstream of the actuation pump and is configured when activated to direct the second flow of fuel toward the main fuel filter when a pressure of the second flow of fuel exceeds a predetermined threshold.

[0023] Additionally or alternatively, in this or other embodiments the actuation pump is a variable displacement pump.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0025] FIG. **1** is a schematic illustration of an embodiment of a gas turbine engine and fuel delivery system through which fuel is delivered to power the gas turbine engine; and

[0026] FIG. **2** is a schematic illustration of an embodiment of a fuel delivery system of a gas turbine engine.

DETAILED DESCRIPTION

[0027] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0028] An aircraft gas turbine engine system is indicated generally at **10** in FIG. **1**. Gas turbine engine system **10** includes a compressor portion **12** operatively coupled to a turbine portion **14** through a shaft **16**. A combustor assembly **18** is fluidically connected between the compressor portion **12** and turbine portion **14**. A fuel delivery system **20** fluidically connects the combustor assembly **14** with a source of fuel **24**. Fuel delivery system **20** may receive fuel directly from source of fuel **24** or, through a compressor stage (not shown) that creates an input pressure for the fuel. In addition to the combustor assembly **18**, the fuel delivery system **20** is operably connected to one or more secondary components, such as an augmentor **26** to deliver a flow of fuel thereto. In some embodiments, a gearbox **28** is operably connected to the gas turbine engine **10** via, for example, the shaft **16** to extract power from the gas turbine engine **10**. The fuel delivery system **20** is operably coupled to the gearbox to power components of the fuel delivery system **20**.

[0029] Referring now to FIG. **2**, the fuel delivery system **20** will now be described in greater detail. The fuel delivery system **20** includes a main pump **30** that directs fuel to the combustor assembly **18** via a combustor fuel line **32**, with a combustor fuel control **34** located along the combustor fuel line **32** to control the flow of fuel to the combustor assembly **18**. Additionally, the fuel delivery system **20** includes an augmentor pump **36** that directs fuel to the augmentor **26** via an augmentor fuel line **38**, with an augmentor fuel control **40** located along the augmentor fuel line **38** to control the flow of fuel to the augmentor **26**. While the present embodiment is described as including an augmentor **36**, one skilled in the art will readily appreciate that this is merely exemplary, and that this arrangement may be utilized to direct fuel to another secondary component of the gas turbine

engine system **10** or the aircraft. The fuel delivery system **20** further includes a boost and actuation pump system **42** utilized to boost fuel pressure of the fuel from the fuel source **24**, and to provide fuel to one or more actuators **44** of the gas turbine engine system **10**. Each of the main pump **30**, the augmentor pump **36** and the boost and actuation pump system **42** are operably connected to the gearbox **28** as a source of power. A flow of fuel is urged from the fuel source **24** and through the boost and actuation pump system **42**, and through one or more of the main pump **30** and the augmentor pump **36** for delivery to one or more of the combustor assembly **18** and the augmentor **26**.

[0030] The boost and actuation pump system **42** is illustrated in FIG. 2. The system **42** includes a boost pump **46** located downstream of the fuel source **24**, but upstream of the main pump **30**. A flow of fuel **50** from the fuel source **24** is directed through the boost pump **46** and through a wash filter **48** downstream of the boost pump **46**. From the wash filter **48**, a first portion **50a** of the flow of fuel is directed to a main fuel filter **52** and toward the main pump **30** and augmentor pump **36**.

[0031] A second portion **50b** of the flow of fuel is diverted from the wash filter **48** along an actuation circuit **54**. The actuation circuit **54** includes an actuation pump **56**, which in some embodiments is a variable displacement pump (VDP). The actuation pump **56** is controlled by a linear variable differential transformer (LVDT) **58** and is coupled to an electrohydraulic servo valve (EHSV) **60**. The actuation pump **56** directs the second portion **50b** toward one or more actuation devices **44** to drive operation of the actuation devices **44**. A check valve (CV) **64** is positioned between the actuation pump **56** and the actuation devices **44**. The check valve **64** is present to protect the system in the event of failure of the actuation pump **56**. The check valve **64** prevents the second portion **50b** from flowing backward across the actuation pump **56** and allows the augmentor pump **36** to take over to support the actuation devices. Further excess second portion **50b** may be directed toward, for example, the augmentor **26** via operation of an augmentor select valve (ASV) **66** located between an actuation filter **68** downstream of the control valve **64**, and the actuation devices **62**. From the actuation devices **44**, the second portion **50b** is circulated to an outlet of the boost pump **46** upstream of the wash filter **48**.

[0032] The actuation circuit **54** includes features to control temperature of the second portion **50b** flowing through the actuation circuit **54**, thus preventing a thermal runaway of the actuation circuit **54**. First, the actuation circuit **54** includes a thermal bypass valve (TBPV) **70** positioned along the actuation circuit **54** downstream of the actuation pump **56**, for example, between the actuation filter **68** and the ASV **66**. In the event of a temperature of the second portion **50b** exceeding a predetermined threshold temperature, the TBPV **70** is opened, a controlled leak path is opened to increase fuel flow across the boost pump **46** and the actuation pump **56** to help reduce the second portion **50b** temperature. Once the second portion **50b** temperature returns to under the threshold, the TBPV **70** may be closed. The TBPV **70** may be actively controlled, or alternatively may be passively thermostatically controlled. Additionally, in some embodiments the actuation circuit **54** includes a high pressure relief valve (HPRV) **72** located downstream of the actuation pump **56** at a location, for example, upstream of the control valve **64**. When a pressure of the second portion **50b** in the actuation circuit **54** exceeds a predetermined threshold pressure, the HPRV **72** opens automatically, diverting the excess second portion **50b** to the main fuel filter **52** and relieving the pressure in the actuation circuit **54**. Once the pressure of the second portion **50b** in the actuation circuit **54** is once again below the predetermined threshold, the HPRV **72** may be closed.

[0033] Placing the actuation pump **56** upstream on the main fuel filter **52** isolates the actuation pump **56** and related components, shielding the actuation circuit **54** from very high fuel temperatures. This allows for utilizing the lighter and more efficient variable displacement pump, and also reduces the fuel temperature to the actuation devices **44**, which extends the service life of the actuation devices **44**, or alternatively allows for a lighter actuation device to be utilized or allows for placing of the actuation devices in a hotter environment.

[0034] The term “about” is intended to include the degree of error associated with measurement of

the particular quantity based upon the equipment available at the time of filing the application. [0035] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0036] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A fuel delivery system for a gas turbine engine, comprising: a fuel source; a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine; a main fuel filter disposed upstream of the main fuel pump; an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine; and a thermal bypass valve located downstream of the actuation pump and configured to be activated when a temperature of the second flow of fuel exceeds a predetermined threshold, to increase a volume of the second flow of fuel through the actuation pump, thereby decreasing the temperature of the second flow of fuel; wherein when the thermal bypass valve is activated the second flow of fuel is directed to the main fuel filter.
2. The fuel delivery system of claim 1, further comprising a boost pump positioned upstream of the main fuel filter and the actuation pump.
3. The fuel delivery system of claim 2, further comprising a wash filter disposed fluidly between the boost pump and the main fuel filter.
4. The fuel delivery system of claim 3, wherein the actuation pump is fed from the wash filter.
5. The fuel delivery system of claim 2, wherein the second flow of fuel is urged from the one or more actuation devices to an outlet of the boost pump.
6. (canceled)
7. The fuel delivery system of claim 1, wherein the thermal bypass valve is one of actively controlled or thermostatically controlled.
8. (canceled)
9. The fuel delivery system of claim 1, further comprising a high pressure relief valve disposed downstream of the actuation pump and configured when activated to direct the second flow of fuel toward the main fuel filter when a pressure of the second flow of fuel exceeds a predetermined threshold.
10. The fuel delivery system of claim 1, wherein the actuation pump is a variable displacement pump.
11. A gas turbine engine and fuel delivery system, comprising: a gas turbine engine, including: a turbine; and a combustor where a first flow of fuel is combusted to drive the turbine via a flow of combustion products; and a fuel delivery system operably connected to the gas turbine engine, including: a fuel source; a main fuel pump configured to deliver a first flow of fuel to the combustor; a main fuel filter disposed upstream of the main fuel pump; an actuation pump disposed

upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine; and a thermal bypass valve located downstream of the actuation pump and configured to be activated when a temperature of the second flow of fuel exceeds a predetermined threshold, to increase a volume of the second flow of fuel through the actuation pump, thereby decreasing the temperature of the second flow of fuel; wherein when the thermal bypass valve is activated the second flow of fuel is directed to the main fuel filter.

12. The gas turbine engine and fuel delivery system of claim 11, further comprising a boost pump positioned upstream of the main fuel filter and the actuation pump.

13. The gas turbine engine and fuel delivery system of claim 12, further comprising a wash filter disposed fluidly between the boost pump and the main fuel filter.

14. The gas turbine engine and fuel delivery system of claim 13, wherein the actuation pump is fed from the wash filter.

15. The gas turbine engine and fuel delivery system of claim 12, wherein the second flow of fuel is urged from the one or more actuation devices to an outlet of the boost pump.

16. (canceled)

17. The gas turbine engine and fuel delivery system of claim 11, wherein the thermal bypass valve is one of actively controlled or thermostatically controlled.

18. (canceled)

19. The gas turbine engine and fuel delivery system of claim 11, further comprising a high pressure relief valve disposed downstream of the actuation pump and configured when activated to direct the second flow of fuel toward the main fuel filter when a pressure of the second flow of fuel exceeds a predetermined threshold.

20. The gas turbine engine and fuel delivery system of claim 11, wherein the actuation pump is a variable displacement pump.

21. A fuel delivery system for a gas turbine engine, comprising: a fuel source; a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine; a main fuel filter disposed upstream of the main fuel pump; an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine; and a thermal bypass valve located downstream of the actuation pump and configured to be activated when a temperature of the second flow of fuel exceeds a predetermined threshold; wherein when the thermal bypass valve is activated the second flow of fuel is directed to the main fuel filter.
